

INTERNSPARK MERN STACK INTERNSHIP

TASK 1: BUILD A RESTFUL API USING NODE.JS, EXPRESS.JS AND MONGODB

Project Name:

InternHub – Internship Management Portal

Submitted By: MD Sohail Ahmed

InternSpark MERN Stack Internship

Objective:

The objective of this project is to develop a RESTful API for an Internship Management Portal using Node.js, Express.js, and MongoDB. The API provides CRUD operations for managing internship records and demonstrates backend development concepts including database integration, request handling, validation, logging, and error management.

The project stores internship records in MongoDB Atlas and exposes endpoints for creating, retrieving, updating, and deleting intern information.

Technologies Used:

Node.js

Express.js

MongoDB Atlas

Mongoose

Morgan

Postman

GitHub

Project Architecture:

internhub-backend/

|

├── config/

├── controllers/

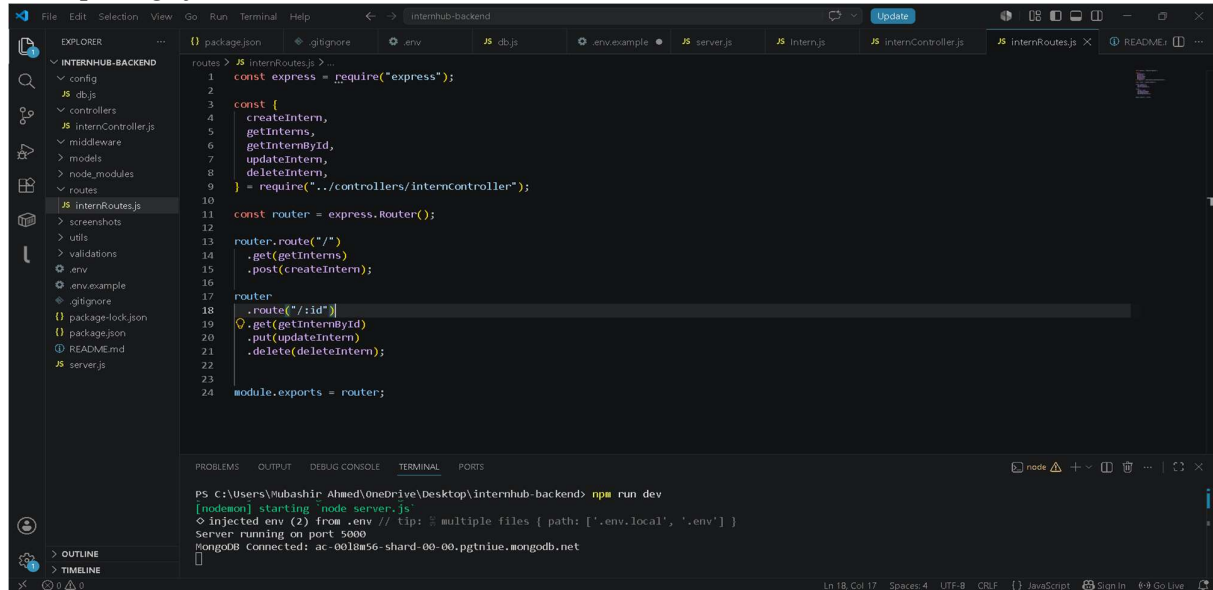
├── middleware/

├── models/

├── routes/

└── postman/

- └─ .env.example
- └─ README.md
- └─ server.js
- └─ package.json



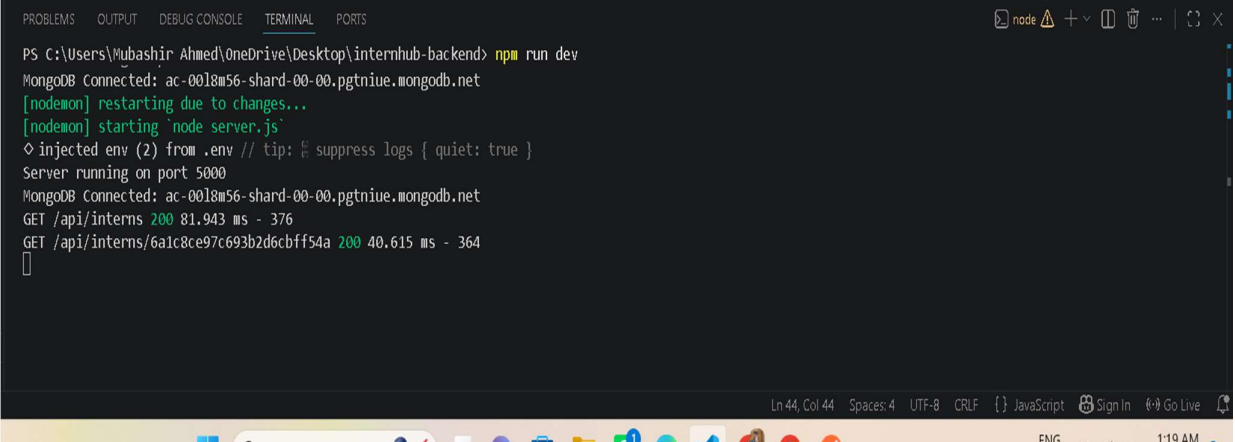
Features:

- Create Intern
- Get All Interns
- Get Single Intern
- Update Intern
- Delete Intern
- MongoDB Atlas Integration
- Mongoose Validation
- Error Handling
- Request Logging using Morgan

API Endpoints:

Method	Endpoint	Purpose
POST	/api/interns	Create Intern
GET	/api/interns	Get All Interns
GET	/api/interns/:id	Get Single Intern
PUT	/api/interns/:id	Update Intern
DELETE	/api/interns/:id	Delete Intern

MongoDB Connection:



```
PS C:\Users\Mubashir Ahmed\OneDrive\Desktop\internhub-backend> npm run dev
MongoDB Connected: ac-0018m56-shard-00-00.pgtniue.mongodb.net
[nodemon] restarting due to changes...
[nodemon] starting 'node server.js'
◇ injected env (2) from .env // tip: ⚙ suppress logs { quiet: true }
Server running on port 5000
MongoDB Connected: ac-0018m56-shard-00-00.pgtniue.mongodb.net
GET /api/interns 200 81.943 ms - 376
GET /api/interns/6a1c8ce97c693b2d6cbff54a 200 40.615 ms - 364
[]
```

MongoDB Atlas was used as the cloud database. The application successfully connects using Mongoose and stores internship records.

CRUD Operations:

Create Intern



```
Body Cookies Headers 8 Test Results 1/1 200 OK • 49 ms • 645 B • Save Respo
{} JSON Preview Visualize
4 "data": [
5   {
6     "_id": "6a1c8ce97c693b2d6cbff54a",
7     "name": "MD Sohail Ahmed",
8     "email": "sohail@example.com",
9     "phone": "9876543210",
10    "college": "XYZ Engineering College",
11    "domain": "MERN Stack",
12    "status": "Active",
13    "startDate": "2026-06-01T00:00:00.000Z",
14    "notes": "First intern record",
15    "createdAt": "2026-05-31T19:32:57.936Z",
16    "updatedAt": "2026-05-31T19:32:57.936Z",
17    "__v": 0
```

Get All Interns

GET Get data

My Collection > Get data

GET http://localhost:5000/api/interns

Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Schema Beautify

```
1 {
2   "status": "Completed",
3   "notes": "Internship completed successfully"
4 }
```

Body Cookies Headers 8 Test Results 1/1

200 OK • 49 ms • 303 B

Save Response

JSON Preview Visualize

```
1 {
2   "success": true,
3   "count": 0,
4   "data": []
5 }
```

Globals Vault Tools

Get Single Intern

GET Get data

My Collection > Get data

GET http://localhost:5000/api/interns/6a1c95b007b8eee344e5e239

Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Schema Beautify

```
1 {
2   "status": "Completed",
3   "notes": "Internship completed successfully"
4 }
```

Body Cookies Headers 8 Test Results 1/1

200 OK • 50 ms • 680 B

Save Response

JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Intern updated successfully",
4   "data": {
5     "_id": "6a1c95b007b8eee344e5e239",
6     "name": "Test Intern",
7     "email": "testintern@example.com",
8     "phone": "9876543211",
9     "college": "XYZ Engineering College",
10    "domain": "MERN Stack",
11    "status": "Completed",
12    "startDate": "2026-06-01T00:00:00.000Z",
13    "notes": "Internship completed successfully",
14    "createdAt": "2026-05-31T20:10:24.897Z",
15    "updatedAt": "2026-05-31T20:12:15.196Z",
16    "__v": 0
17  }
18 }
```

Globals Vault Tools

Update Intern

PUT Get data No environment

My Collection > Get data Save Share

PUT http://localhost:5000/api/interns/6a1c8ce97c693b2d6cbff54a Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "status": "Completed",
3   "notes": "Internship completed successfully"
4 }
```

Body Cookies Headers 8 Test Results 1/1

200 OK • 680 ms • 690 B Save Response

{ JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Intern updated successfully",
4   "data": {
5     "_id": "6a1c8ce97c693b2d6cbff54a",
6     "name": "MD Sohail Ahmed",
7     "email": "sohail@example.com",
8     "phone": "9876543210",
9     "college": "XYZ Engineering College",
10    "domain": "MERN Stack",
11    "status": "Completed",
12    "startDate": "2026-06-01T00:00:00.000Z",
13    "notes": "Internship completed successfully",
14    "createdAt": "2026-05-31T19:32:57.936Z",
15  }
16 }
```

Globals Vault Tools

Delete Intern

My Collection > Get data Save Share

DELETE http://localhost:5000/api/interns/6a1c93f607b8eee344e5e238 Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "name": "Test Intern",
3   "email": "testintern@example.com",
4   "phone": "9876543211",
5   "college": "XYZ Engineering College",
6   "domain": "MERN Stack",
7   "status": "Active",
8   "startDate": "2026-06-01",
9   "notes": "For documentation screenshots"
10 }
```

Body Cookies Headers 8 Test Results 1/1

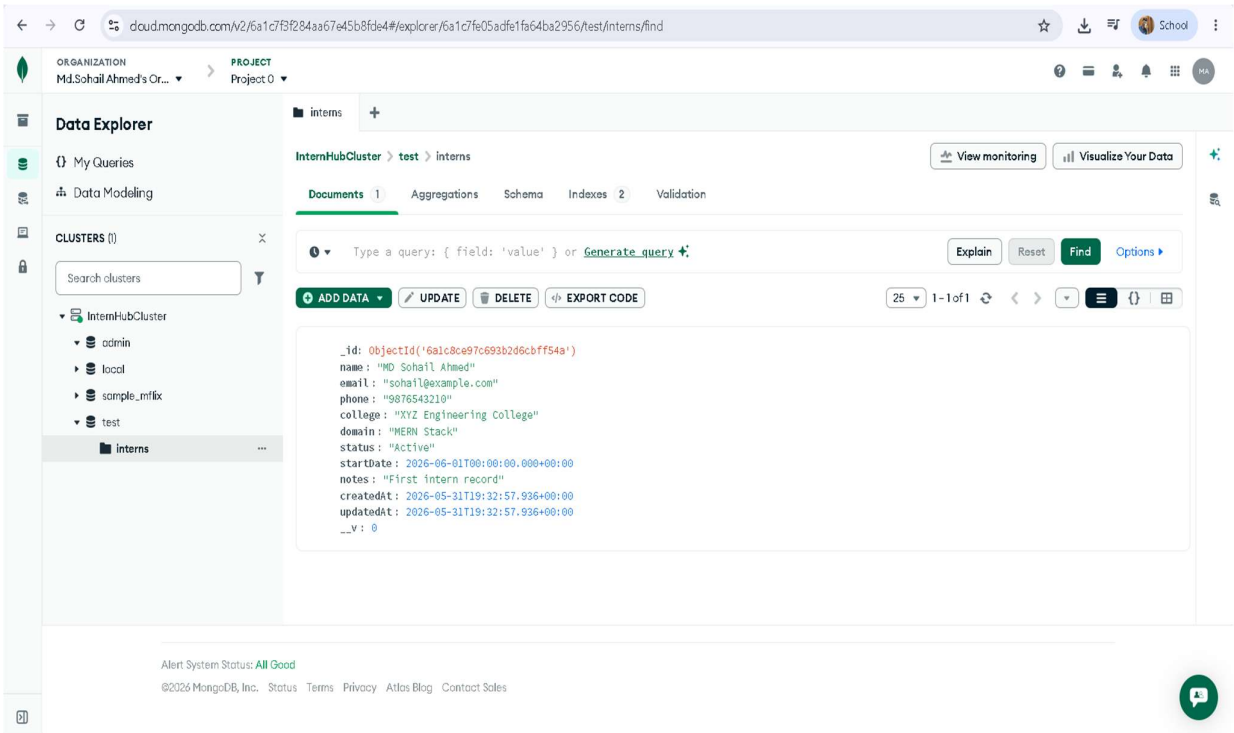
200 OK • 49 ms • 323 B Save Response

{ JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Intern deleted successfully"
4 }
```

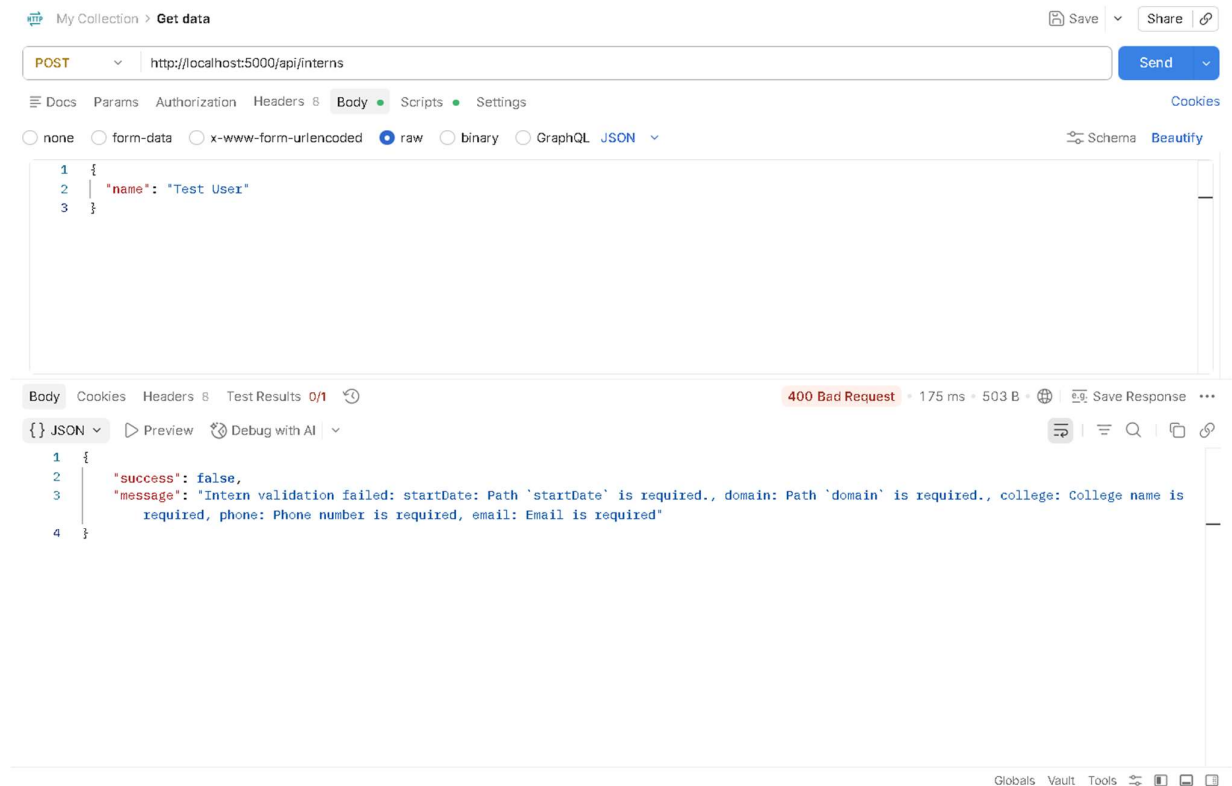
Globals Vault Tools

MongoDB Data Storage



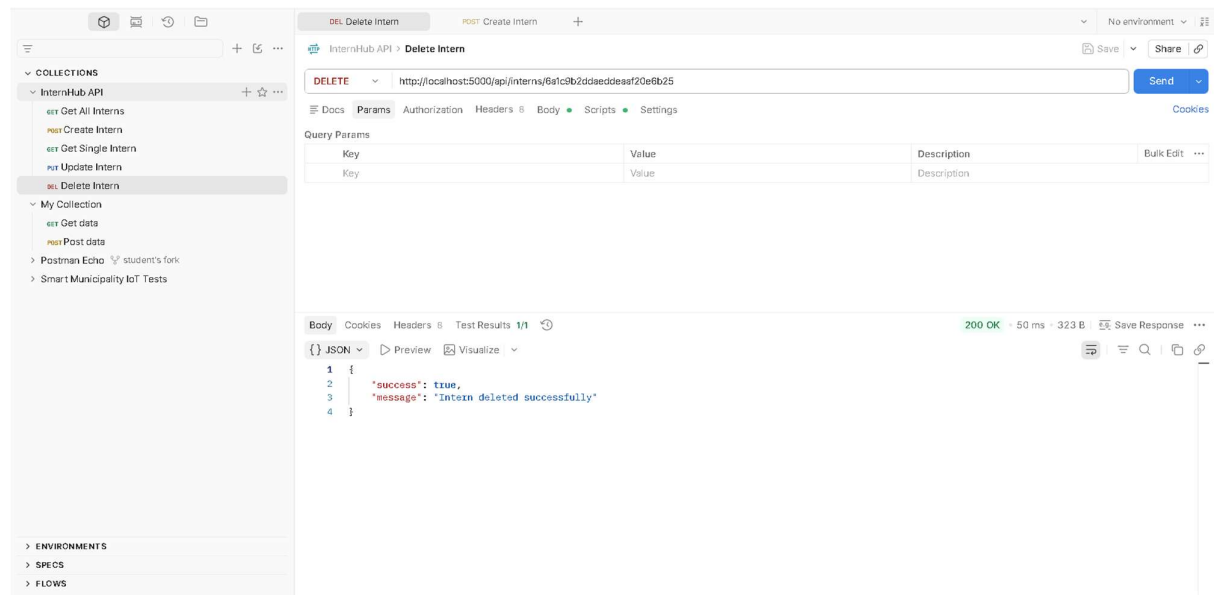
The API successfully stores intern records in MongoDB Atlas and retrieves them through REST endpoints.

Validation and Error Handling



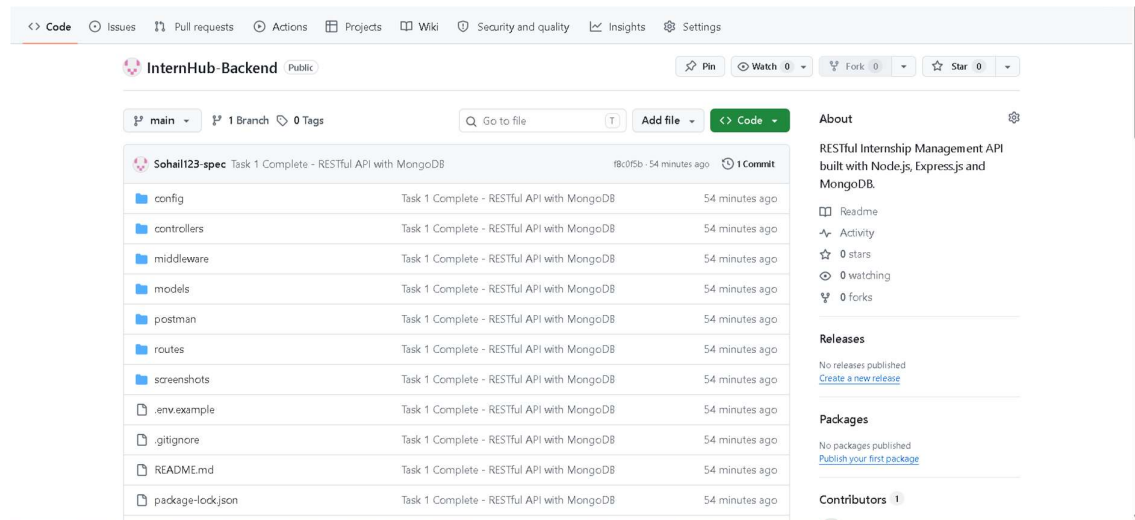
Mongoose schema validation ensures that required fields are provided before storing records. Invalid requests are rejected with appropriate error messages.

Postman Collection



All CRUD API requests were organized into a Postman Collection and exported for testing and documentation purposes.

GitHub Repository



GitHub Repository Link: <https://github.com/Sohail123-spec/InternHub-Backend>

Conclusion:

The Internship Management Portal backend was successfully developed using Node.js, Express.js, and MongoDB. The application provides complete CRUD functionality, database integration, validation, logging, and error handling. The project satisfies all requirements specified in Task 1 and establishes a strong backend foundation for future frontend integration and authentication features.

Submitted By: MD Sohail Ahmed

Submission Date: 01/06/2026